Md. Saif Hassan Onim
UT-ORII GATE Fellow | Ph.D. Candidate, Computer Engineering, University of Tennessee, Knoxville, TN
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Research Summary

Research focused on machine learning, quantum machine learning, and digital health analytics, with publications spanning IEEE, ACM, and biomedical computing venues. Work integrates physiological sensing, time-series modeling, and hybrid quantum-classical algorithms for stress, emotion, and Alzheimer's-related biomarker detection. Experienced in building end-to-end research pipelines from sensor data acquisition and processing to model deployment on embedded and FPGA platforms and in applying QML techniques to healthcare, cybersecurity, and cyber-physical systems. Research contributions include interdisciplinary collaborations with clinical teams, development of reproducible HPC workflows, and award-winning work on quantum-hybrid anomaly detection and VLSI-oriented ML systems.

Education

Ph.D., Computer Engineering - University of Tennessee, Knoxville (2022-Present)
GPA: 4.00/4.00 | Graduation: 2026 (Expected)

M.Sc., Computer Engineering - University of Tennessee, Knoxville (2022-2024)
GPA: 4.00/4.00 | Graduation: Dec 2024

B.Sc., Electrical, Electronic & Communication Engineering - MIST, Bangladesh (2016-2020)
GPA: 3.72/4.00 | Graduation: Dec 2019

Research Interests

Machine Learning, Quantum Machine Learning, Adversarial AI, Computer Vision, Deep Learning, Image Segmentation, TinyML, Stress Detection, Wearable Devices, Embedded Systems

Key Skills

Programming Languages: Python, C, C++, Verilog/SystemVerilog, VHDL, Matlab, ARM/x86 Assembly.
Design Tools: Virtuoso, Vivado, Innovus, Keil (AMD), Quartus.
Hardware: RTOS, FPGA Design and Synthesis.
Machine Learning & Quantum Libraries: Keras, Pytorch, Tensorflow, IBM Qiskit, OpenQASM, Pennylane, Intel QSDK, Nvidia CudaQ
Operating System: Windows, Linux (Kali, Ubuntu).

Research & Publications

Posters:

- **Md. Saif Hassan Onim** and Himanshu Thapliyal, "Exploring the Feasibility of Wearable Sensors for Emotion Detection of Older Adults," *Alzheimer's Association International Conference, Toronto, Canada, July 27-31, 2025*
- Brian G. Carter, **Md. Saif Hassan Onim**, Nancy Wolff, Himanshu Thapliyal and Elizabeth K. Rhodus, "Analyzing Stress Reactions in Community-Dwelling Individuals with Alzheimer's Disease and Associated Behavioral Symptoms," *The 20th Annual Spring Conference of the UK Center for Clinical and Translational Science (CCTS), 2025*

- Brian G. Carter, **Md. Saif Hassan Onim**, Nancy Wolff, Himanshu Thapliyal and Elizabeth K. Rhodus, “Assessment of physiological and observable stress responses during functional tasks in community-dwelling individuals with Alzheimer’s disease and associated behavioral symptoms,” *AOA Groves Memorial Student Research Symposium, 2025*
- **Md. Saif Hassan Onim** and Himanshu Thapliyal, “Performance of Quantum Hybrid Support Vector Machine in Stress Detection,” *Quantum Computing User Forum, 2024, ORNL, Knoxville, TN, USA, August 12-15, 2024*
- Elizabeth K. Rhodus, **Md. Saif Hassan Onim**, Celeste Roberts, Sanjeev Kumar, Amer M Burhan, and Himanshu Thapliyal, “Utilization of Wearable Devices as Means for Remote Digital Biometric Data Collection in a Community-based, Rural Randomized Controlled Trial Among Alzheimer’s Disease Dyads,” *in Alzheimer’s Association International Conference. ALZ, 2024*
- Himanshu Thapliyal and **Md. Saif Hassan Onim**,“Predicting Psychological and Physiological Stress with Biomarkers and Wearable Sensors”, *Military Health System Research Symposium (MHSRS), 2024*
- Celeste Roberts, **Md. Saif Hassan Onim**, Sanjeev Kumar, Amer M Burhan, Himanshu Thapliyal and Elizabeth K. Rhodus, “Implementation of RCT Intervention with Wearable Digital Biometric Collection in Rural Community-residing Dementia Dyads”, *The 19th Annual Spring Conference of the UK Center for Clinical and Translational Science (CCTS), 2024*

Conference Papers:

- **Md. Saif Hassan Onim**, Travis S. Humble and Himanshu Thapliyal, “Emotion Recognition in Older Adults with Quantum Machine Learning and Wearable Sensors”, *Proceedings of IEEE Computer Society Annual Symposium on VLSI 2025 (ISVLSI ’25), Kalamata, Greece, July 6-9, 2025*
- **Md. Saif Hassan Onim** and Himanshu Thapliyal, “Emotion Detection in Older Adults Using Physiological Signals from Wearable Sensors”, *Proceedings of the Great Lakes Symposium on VLSI 2025 (GLSVLSI 2025), New Orleans, LA, USA. ACM, New York, NY, USA, June 30-July 2, 2025*
- **Md. Saif Hassan Onim**, Travis S. Humble and Himanshu Thapliyal, “Quantum Hybrid Support Vector Machines for Stress Detection in Older Adults”, *IEEE 43rd International Conference on Consumer Electronics (ICCE 2025), Las Vegas, NV, USA, 2025*
- **Md. Saif Hassan Onim** and Himanshu Thapliyal, “Detection of Physiological Data Tampering Attacks with Quantum Machine Learning”, *2025 IEEE International Symposium on Circuits and Systems (ISCAS 2025), London, United Kingdom, 2025 (accepted)*
- Tyler Cultice, **Md. Saif Hassan Onim**, Annarita Giani and Himanshu Thapliyal, “Anomaly Detection for Real-World Cyber-Physical Security using Quantum Hybrid Support Vector Machine”, *Proceedings of IEEE Computer Society Annual Symposium on VLSI 2024 (ISVLSI 2024), Knoxville, TN, USA, July 1-3, 2024 (Best Paper Award)*
- **Md. Saif Hassan Onim** and Himanshu Thapliyal, “Predicting Stress in Older Adults with RNN and LSTM from Time Series Sensor Data and Cortisol”, P*roceedings of IEEE Computer Society Annual Symposium on VLSI 2024 (ISVLSI 2024), Knoxville, TN, USA, July 1-3, 2024*
- **Md. Saif Hassan Onim** and Himanshu Thapliyal, “CASD-OA: Context-Aware Stress Detection for Older Adults with Machine Learning and Cortisol Biomarker”, *Proceedings of the Great Lakes Symposium on VLSI 2023 (GLSVLSI 2023), Knoxville, TN, USA. ACM, New York, NY, USA, June 5-7, 2023*
- Md. Akiful Hoque Akif, Koushik Roy, Nawsher Abdullah, Nusrat Zahan Priota and **Md. Saif Hassan Onim**, “Performance Analysis of Machine Learning Models for Cheating Detection in Online Examinations”, *Proceedings of IEEE 25th International Conference on Computer and Information Technology (ICCIT), Cox’s Bazar, Bangladesh, 2022*
- Md Wahiduzzaman Arnob, Arunima Dey Pooja and **Md. Saif Hassan Onim**, “A Combined PCA-MLP Network for Early Breast Cancer Detection”, *Proceedings of 2022 IEEE 2ND Asian Conference on Innovation in Technology (ASIANCON2022), Pune, India, August 2022*
- Tareque Bashar Ovi, Sauda Suara Naba, Dibaloke Chanda and **Md. Saif Hassan Onim**, “A Transfer-Learning Based Ensemble Architecture for ECG Signal Classification”, *Proceedings of IEEE Region 10 Symposium (TENSYMP2022), Mumbai, India, Jun. 2022*
- Abtahi Ishmam, Mahmudul Hasan, **Md. Saif Hassan Onim**, Koushik Roy, Md. Akiful Hoque Akif and Hussain Nyeem, “Modelling Lips-State Detection Using CNN for Non-Verbal Communications”, *Proceedings of IEEE International Conference on 4th Industrial Revolution and Beyond (IC4IR), 2021, Dhaka, Bangladesh, 2021*

- Shah Mohazzem Hossain, **Md. Saif Hassan Onim**, Sagor Biswas and Abdul Hasib Chowdhury, "Application of Machine Learning for Optimal Placement of Distributed Generation", *Proceedings of IEEE International Conference on Information and Communication Technology for Sustainable Development (ICICT4SD), Dhaka, Bangladesh, pp. 231-234. 2021*
- **Md. Saif Hassan Onim**, Muhaiminul Islam Akash, Mahmudul Haque and Raiyan Ibne Hafiz, "Traffic Surveillance using Vehicle License Plate Detection and Recognition in Bangladesh", P*roceedings of IEEE International Conference on Electrical and Computer Engineering, pp. 121-124, 2020*
- **Md. Saif Hassan Onim**, Aiman Rafeed Ehtesham, Amreen Anbar, A. K. M. Nazrul Islam and A. K. M. Mahbubur Rahman, "LULC Classification by Semantic Segmentation of Satellite Images Using FastFCN", *Proceedings of IEEE International Conference on Advanced Information and Communication Technology, pp. 472-476, 2020*

Journals:

- Tyler Cultice, **Md. Saif Hassan Onim**, Himanshu Thapliyal and Annaria Giani, "Investigating Quantum-Hybrid Support Vector Machines for Anomaly Detection in Industrial Control Systems" *IEEE Transactions on Emerging Topics in Computing, 2025 (submitted)*
- **Md. Saif Hassan Onim**, Himanshu Thapliyal and Elizabeth K. Rhodus, "Utilizing Machine Learning for Context-Aware Digital Biomarker of Stress in Older Adults" *Information, 15, no. 5: 274, 2024*
- Dibaloke Chanda, **Md. Saif Hassan Onim**, Hussain Nyeem, Tareque Bashar Ovi, and Sauda Suara Naba, "Dcensnet: A New Deep Convolutional Ensemble Network for Skin Cancer Classification", *Biomedical Signal Processing and Control, vol. 89, p. 105757, 2024*
- **Md. Saif Hassan Onim**, Himanshu Thapliyal and Elizabeth K. Rhodus "A Review of Context-Aware Machine Learning for Stress Detection", *IEEE Consumer Electronics Magazine, vol. 13, no. 4, pp. 10-16, July 2024*
- **Md. Saif Hassan Onim**, Hussain Nyeem, Md. Wahiduzzaman Khan Arnob and Arunima Dey Pooja "Unleashing the power of generative adversarial networks: A novel machine learning approach for vehicle detection and localisation in the dark," *Cognitive Computation and Systems, vol. 5, no. 3, pp. 169-180, 2023.*
- **Md. Saif Hassan Onim**, Zubayar Mahatab Md Sakif, Adil Ahnaf, Ahsan Kabir, Abul Kalam Azad, Amanullah Maung Than Oo, Rafina Afreen, Sumaita Tanjim Hridy, Mahtab Hossain, Taskeed Jabid and Md Sawkat Ali, "Solnet: A Convolutional Neural Network for Detecting Dust on Solar Panels", *Energies, vol. 16, no. 1, 2023*
- **Md. Saif Hassan Onim**, Hussain Nyeem, Koushik Roy, Mahmudul Hasan, Abtahi Ishmam, Md. Akiful Hoque Akif, Tareque Bashar Ovi, "BLPnet: A new DNN model and Bengali OCR engine for Automatic License Plate Recognition", *Array, vol. 15, p. 100244, 2022*

Research and Teaching Experience

Graduate Research and Teaching Assistant, Dept of EECS, University of Tennessee, Knoxville (August, 2022-Present)

- Assisted in teaching undergraduate and graduate courses in Computer Engineering and Machine Learning, supporting instruction through labs, grading, and development of programming and hardware-based materials. Mentored students individually and in groups to strengthen conceptual understanding. Conducted research in machine learning, quantum machine learning, and digital health analytics, including development of data pipelines, hybrid models, and FPGA-based implementations for stress, emotion, and Alzheimer's detection. Contributed to interdisciplinary projects, managed HPC workflows, and co-authored publications and technical reports advancing computational methods in health-related applications.

Adjunct Lecturer & Research Mentor - MIST, Bangladesh *(2020-2022)*

- Served as an adjunct lecturer delivering lectures, evaluating student work, updating course materials, and contributing to course file preparation. Conducted short courses on Python, introductory programming, and image and video processing to strengthen foundational and applied skills among undergraduates. Supported senior faculty by assisting in undergraduate thesis supervision and providing academic guidance to students across multiple research projects. Mentored undergraduate researchers on independent studies, scholarly writing, and effective literature review practices to enhance research quality and productivity.

Awards & Distinctions

- **2025:** Web Chair, 19th IEEE Dallas Circuits and Systems Conference
- **2025:** Web Chair, IEEE Dallas CAS section
- **2025-present:** Graduate Advancement, Training and Education (GATE) Fellow at The University of Tennessee-Oak Ridge Innovation Institute (UT-ORII).
- **2024-2025:** Senator, Graduate Student Senate, UTK
- **2023-2024:** Gonzalez Family Award for Outstanding Graduate Teaching Assistant in Computer Engineering, EECS, UTK
- **2023-2024:** Publicity and Social Media Officer, Bangladesh Student Association (BSA), University of Tennessee, Knoxville, USA
- **2024:** Student Activities Coordinator, IEEE East Tennessee Section
- **2024:** Presentation Co-Chair, IEEE Computer Society Annual Symposium on VLSI
- **2021-present:** Graduate Student Member, Institute of Electrical and Electronics Engineers (IEEE).
- **2023-present:** Member, Association for Computing Machinery (ACM).
- **2019:** MIST Merit Scholarship
- **2018:** MIST Dean's List Award

References

Dr. Himanshu Thapliyal
Inaugural Walden and Paula Rhines Endowed Quantum Informatics Professor
Lyle School of Engineering, Southern Methodist University (SMU), Dallas, Texas, USA
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